A Mechanism-Informed Gene Signature for Regenerative vs. Fibrotic Tendon Healing

Mapping the transcription-factor program behind the fate decision in tendon progenitors — and compressing it into a validated 23-gene Healing Index.

Fangyi Li[1] · Wilder Scott[2] · Yan Yan[3]

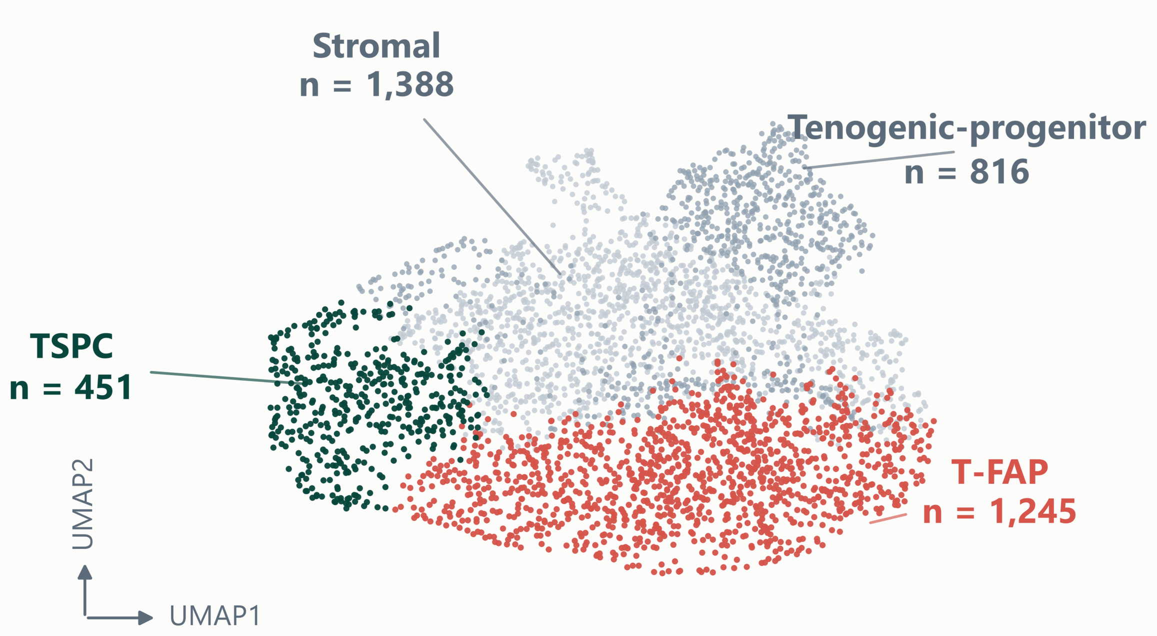
[1]Department of Integrative Biology, College of Biological Science, University of Guelph · [2]Sunnybrook Research Institute · [3]School of Computer Science, College of Computational, Mathematical and Physical Sciences, University of Guelph

Sunnybrook

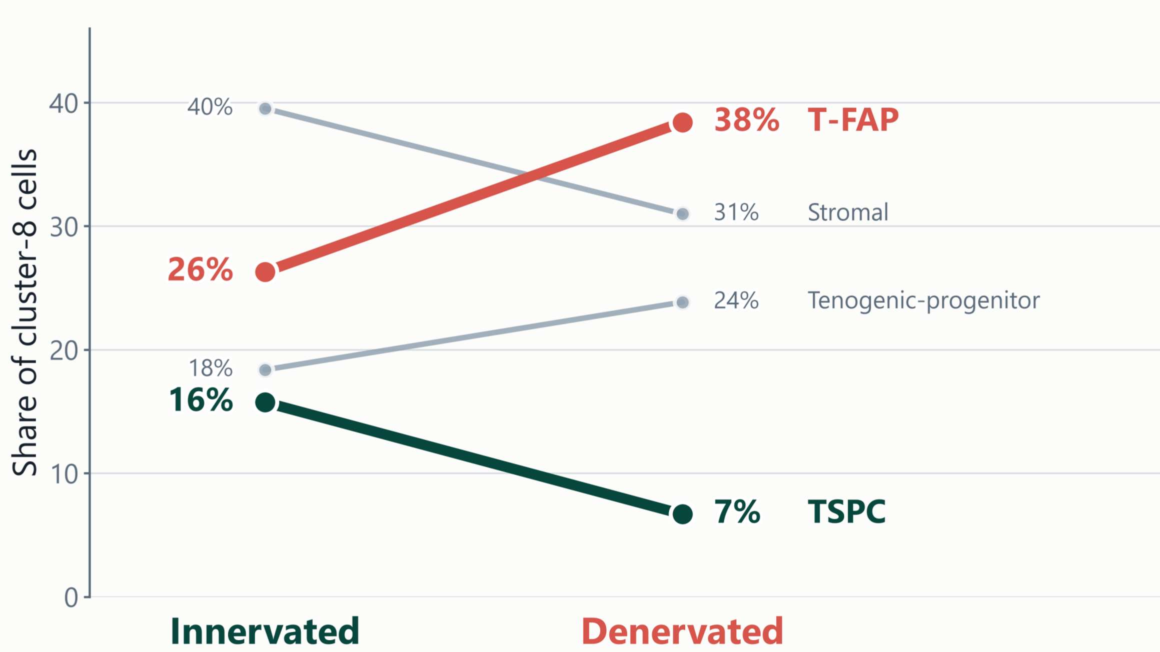
UNIVERSITY OF GUELPH · College of Biological Science · Department of Integrative Biology

1 Introduction

Tendon heals badly: repair tissue is scar, not aligned tendon. Two progenitor populations inside the same PDGFRα⁺ pool decide which one forms — tendon stem/progenitor cells (TSPC) rebuild tendon, tendon fibro-adipogenic progenitors (T-FAP) lay down scar (Harvey et al., 2019).



3,900 PDGFRα⁺ cells from injured mouse tendon.
Sub-clustering resolves four populations (Cherief et al., 2023; day 14 post-injury).



Blocking sensory-nerve signalling shifts the pool.
TSPC share falls by more than half; T-FAP becomes the largest population. Nerve signalling is blocked chemical-genetically (TrkA-F592A mice + 1NMPP1), not surgically — "denervated" is shorthand throughout.

OBJECTIVE

Identify the transcription-factor program that carries out this innervation-dependent fate decision, then compress it into a minimal, validated gene signature that scores how regeneratively a tendon is healing.

2 Methods

Two independent public mouse scRNA-seq datasets, analysed through one integrated pipeline. No new animal work was performed.

Harvey et al. (2019)
Patellar tendon, uninjured · 4,069 cells after QC · SRA PRJNA506218
Ground-truth TSPC / T-FAP labels. Training set.

Cherief et al. (2023)
Achilles tendon, day 14 post-injury · 22,615 cells · GEO GSE244921
Innervated vs. TrkA-blocked. Held-out test set.

Analysis pipeline

1. **QC, clustering, annotation**
   PDGFRα⁺ pool (3,900 cells) sub-clustered into four populations
2. **pySCENIC — TF regulon inference**
   Run independently per dataset, then compared (Van de Sande et al., 2020)
3. **CellRank 2 — fate drivers**
   Genes correlated with commitment to each terminal population (Weiler et al., 2024)
4. **LIANA — cell–cell communication**
   Ligand–receptor signalling, innervated vs. blocked (Dimitrov et al., 2022)
5. **Feature selection**
   LASSO, Boruta and XGBoost run separately on 124 mechanism-derived candidates
6. **Classifier and Healing Index**
   Logistic regression trained on Harvey, applied unchanged to Cherief

Robustness checks

Every headline claim was re-tested. Two did not survive.

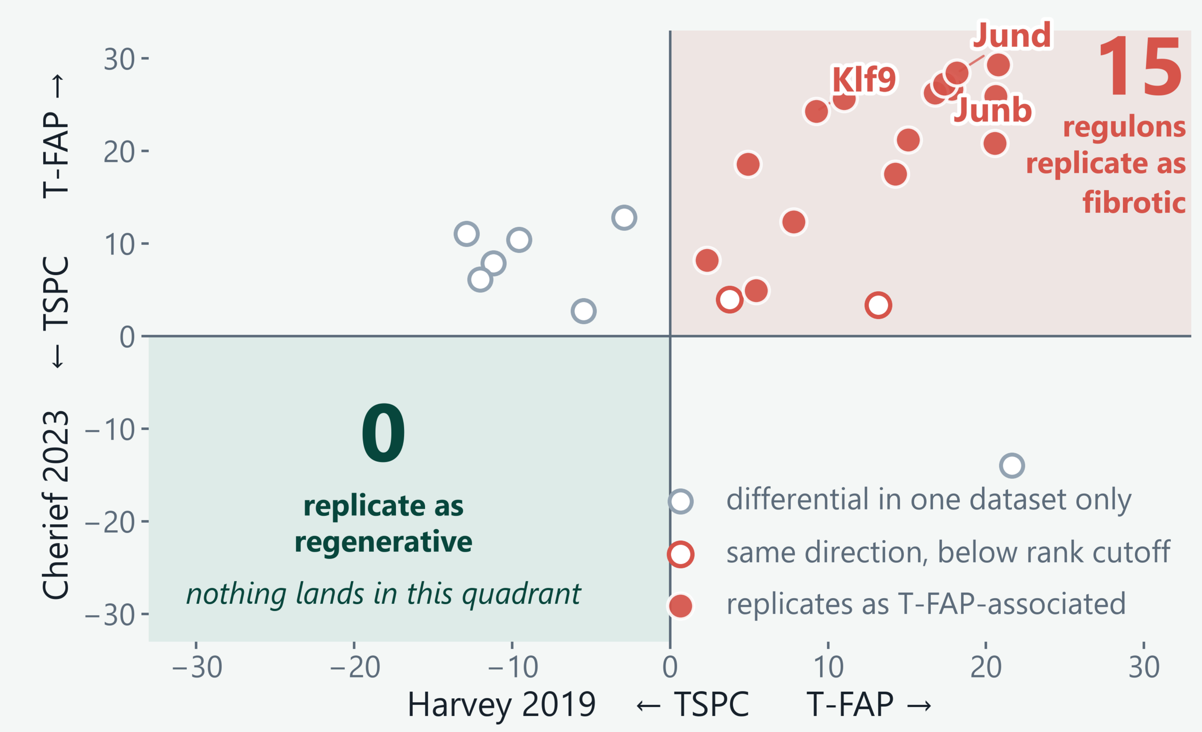
"Efna1→Epha3 is the strongest signal"
LIANA permutation test, n = 100
RETRACTED · p = 1.0

"T-FAP regulons feed back on Pdgfra"
Hypergeometric, BH-corrected
RETRACTED · nothing at $p < 0.05$

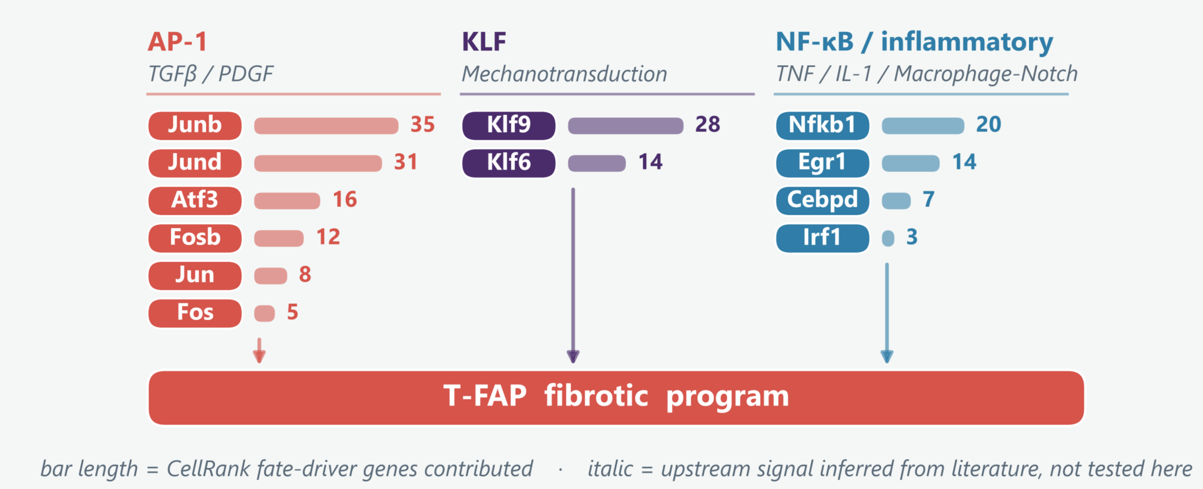
"Zero TSPC regulons replicate"
200× power-matched stability selection
CONFIRMED · still zero at matched n

3 Results

Which transcription-factor regulons separate the two fates, and do they replicate in a second, independent dataset?



Fifteen regulons replicate as fibrotic. None replicate as regenerative.
A 200-iteration power-matched audit reproduced the empty quadrant, so it is not a sample-size artefact.

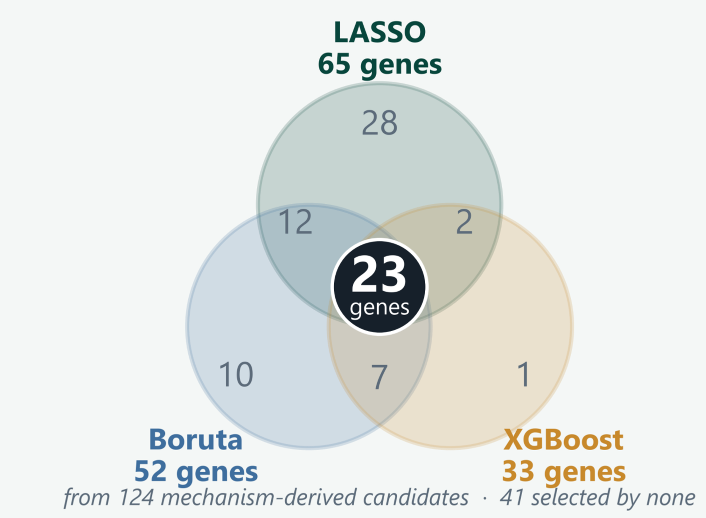


The surviving program is an AP-1 / KLF / NF-κB network.
Junb, Jund and Klf9 contribute the most fate-driver genes; 14 of the 23 final panel genes are their targets. 12 of the 15 regulons are shown — three contribute none.

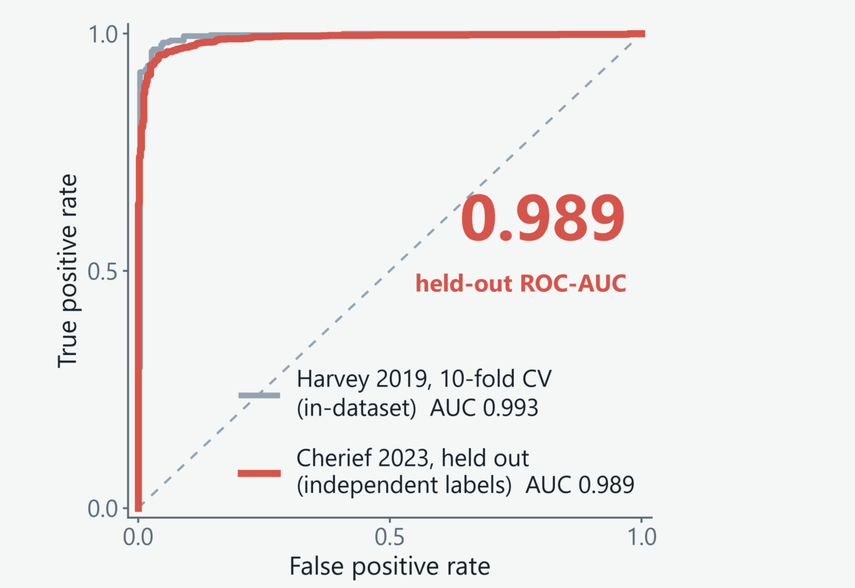
KEY FINDING

The fate decision is asymmetric. Innervation appears to suppress a default fibrotic program rather than to activate a regenerative one — the mirror image of our starting hypothesis.

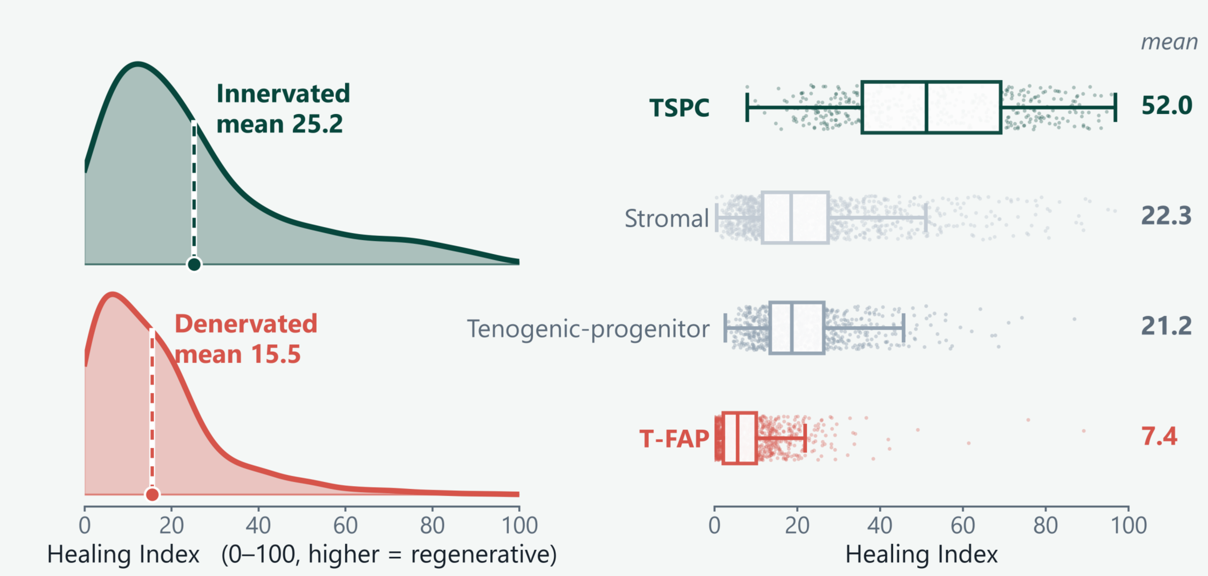
From mechanism to a gene signature



Three methods agree on 23 genes.
LASSO, Boruta and XGBoost ran independently on 124 mechanism-derived candidates.



The panel generalises to a held-out dataset.
Trained on Harvey et al. (2019), then applied unchanged to Cherief et al. (2023) — independently labelled, never seen during selection or training.



The Healing Index tracks innervation status.
It also places two cell identities it never saw in training (Stromal, Tenogenic-progenitor) between the two fates.

4 Conclusions

A mechanism-derived 23-gene Healing Index now tracks the regenerative-to-fibrotic shift across two independent tendon datasets.

- Innervation suppresses a fibrotic program rather than activating a regenerative one: 15 T-FAP regulons replicate across datasets, 0 TSPC regulons do.
- A 23-gene panel separates the two fates at 0.989 ROC-AUC on a held-out, independently labelled dataset.
- The Healing Index falls from 25.2 to 15.5 when nerve signalling is blocked, and orders unseen cell identities sensibly.

LIMITATIONS · WHAT'S NEXT

- No reproducible regenerative regulon is not proof none exists — both datasets are small.
- Neither dataset has biological replicates — validation is cross-dataset, not cross-animal.
- TSPC recall is 77–78% vs. 98–99% for T-FAP: the index flags fibrotic drift better than it confirms repair.
- Next: score newly generated lab tendon scRNA-seq — the first true biological-replicate test.

Data & code availability
All data are publicly available (GEO GSE244921; SRA PRJNA506218). No unpublished or confidential data were used. Computation was performed on the Digital Research Alliance of Canada (NIBI). Code and analysis notebooks: github.com/lifangy6/sunnybrook-tendon-fate-mapping

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